
LAB 2

Logic Minimisation & Universal Gates

From truth tables to minimal circuits --- practice with the solution directly beside the problem.

35 Questions + Detailed Worked Solutions

Computer Architecture & Digital Logic • GATE Preparation

PRINT-READY LATEX EDITION

SECTION A

Warm-up & Fundamentals

Basic • Q1--Q7

STRATEGY — Canonical Forms

- SOP: 1 → uncomplemented, 0 → complemented
- POS: 1 → complemented, 0 → uncomplemented
- Minterm number = binary value of the input combination (MSB left)
- Maxterm number = same binary; product of all maxterms where $F = 0$

Q1

Basic

2 marks - Evaluation

Evaluate $F = A \cdot B + A \cdot B' + A' \cdot B$ for $A=1$, $B=0$. What is F ?

WORKED SOLUTION

STEP 1 — Substitute

$A=1$, $B=0 \rightarrow B'=1$

STEP 2 — Terms

$A \cdot B=0$, $A \cdot B'=1$, $A' \cdot B=0 \rightarrow F=1$

FINAL ANSWER $\rightarrow F = 1$

Q2

Basic

3 marks - Truth Table $\rightarrow$ SOP

Write the canonical SOP for $F(A,B,C)$ from the truth table:

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

WORKED SOLUTION**STEP 1 — Rows where F=1**

(0,0,1), (0,1,0), (1,0,0), (1,1,1)

STEP 2 — Minterms $m_1=A'B'C$, $m_2=A'BC'$, $m_4=AB'C'$, $m_7=ABC$ **FINAL ANSWER** $\rightarrow F = A'B'C + A'BC' + AB'C' + ABC = \Sigma m(1,2,4,7)$ **Trap:** For SOP a 0 must produce a complemented literal.**Q3**

Basic

3 marks - Truth Table $\rightarrow$ POS

Using the same truth table as Q2, write the canonical POS for F.

WORKED SOLUTION**STEP 1 — Rows where F=0**(0,0,0), (0,1,1), (1,0,1), (1,1,0) $\rightarrow M_0, M_3, M_5, M_6$ **STEP 2 — Maxterms** $M_0=(A+B+C)$, $M_3=(A+B'+C')$, $M_5=(A'+B+C')$, $M_6=(A'+B'+C)$ **FINAL ANSWER** $\rightarrow F = (A+B+C)(A+B'+C')(A'+B+C')(A'+B'+C) = \Pi M(0,3,5,6)$ **Q4**

Basic

2 marks - $\Sigma \rightarrow \Pi$ Convert $F(A,B,C) = \Sigma m(0,3,5,6)$ into Π notation.**WORKED SOLUTION**Missing minterms ($F=0$) are 1,2,4,7 $\rightarrow$ maxterms.**FINAL ANSWER** $\rightarrow F = \Pi M(1,2,4,7)$ **Q5**

Basic

2 marks - Minterm ID

What is the minterm for $A=1$, $B=0$, $C=1$, $D=0$? Give algebraic form and decimal index.**WORKED SOLUTION**Binary $1010_2 = 10_{10} \rightarrow A B' C D'$ **FINAL ANSWER** $\rightarrow m_{10} = A B' C D'$

Q6

Basic

2 marks - Maxterm ID

Write the maxterm for $A=0, B=1, C=1$. Also give its decimal index.

WORKED SOLUTION**STEP 1 — Binary**

$$011_2 = 3_{10}$$

STEP 2 — Maxterm rule

$$M_3 = (A + B' + C')$$

FINAL ANSWER → $M_3 = (A + B' + C')$ **Trap:** Students often apply the SOP complement rule to maxterms.**Q7**

Basic

3 marks - Dual Conversion

Given $F = \Sigma m(1,2,4,7)$, write both canonical SOP algebraic expression and equivalent canonical POS.

WORKED SOLUTION

$$\text{SOP: } A'B'C + A'BC' + AB'C' + ABC$$

Missing → maxterms 0,3,5,6

$$\text{POS: } (A+B+C)(A+B'+C')(A'+B+C')(A'+B'+C)$$

FINAL ANSWER → $\text{SOP} = \Sigma m(1,2,4,7)$
 $\text{POS} = \Pi M(0,3,5,6)$

SECTION B

Boolean Algebra & Simplification

Basic → Intermediate • Q8--Q13

STRATEGY — Algebraic Minimisation

- Apply identity, null, complement, idempotent, absorption, distributive laws
- Factor or create cancelling terms; use consensus when useful
- Verify by substitution or truth-table check
- Count gates / literals for area–power–speed impact

Q8

Basic

2 marks - Law ID

Which law justifies $X + X \cdot Y = X$?

A. Distributive B. Absorption C. Idempotent D. DeMorgan

WORKED SOLUTION

$X + XY = X(1+Y) = X \cdot 1 = X$ (Absorption)

FINAL ANSWER → B. Absorption

Q9

Basic

2 marks - Law Application

Simplify $A + A'B$. Name the law used.

WORKED SOLUTION

$A + A'B = (A+A')(A+B) = 1 \cdot (A+B) = A+B$ (Distributive / cover)

FINAL ANSWER → A + B

Q10

Intermediate

3 marks - Algebraic Min

Minimise $F = XY + X'Z + YZ$. Is the YZ term necessary?

WORKED SOLUTION

By consensus theorem, YZ is the consensus of XY and $X'Z \rightarrow$ redundant.

FINAL ANSWER → $F = XY + X'Z$ (YZ can be dropped)

Q11

Intermediate

3 marks - Equivalence

Prove or disprove: $(A+B)(A'+C) = AC + A'B$.

WORKED SOLUTION

Expand LHS: $AA' + AC + A'B + BC = AC + A'B + BC$. Consensus removes BC → $AC + A'B$. Equality holds after dropping consensus.

FINAL ANSWER → True (after consensus); $F = AC + A'B$

Q12

Intermediate

3 marks - Redundant Terms

Identify and remove all redundant terms from $F = A'B'C + A'BC + AB'C + ABC + A'B$.

WORKED SOLUTION

$A'B'C + A'BC = A'C$; $AB'C + ABC = AC$; then $A'C + AC = C$; remaining $A'B$. Result: $F = C + A'B$

FINAL ANSWER → $F = C + A'B$

Q13

Intermediate

3 marks - Dual / Complement

Find the complement of $F = (A+B)(A'+C)$ and simplify.

WORKED SOLUTION

$F' = (A+B)' + (A'+C)' = A'B' + AC'$ (DeMorgan)

FINAL ANSWER → $F' = A'B' + AC'$

SECTION C

K-map Foundations

Intermediate • Q14--Q20

K-MAP STRATEGY

- Fill correctly (Gray-code order)
- Largest groups only (powers of 2: 1,2,4,8...)
- Rectangular & adjacent (incl. wrap-around)
- Every 1 covered; overlapping allowed
- Variables that change inside a group are eliminated

Q14

Intermediate

3 marks - Gray Code

Why must K-map order be 00-01-11-10 instead of natural binary 00-01-10-11?

WORKED SOLUTION

Adjacent cells must differ by exactly one bit for the combining theorem. Natural order places 10 next to 01 (two-bit difference). Gray code guarantees single-bit adjacency including wrap-around.

FINAL ANSWER → Gray code ensures single-bit adjacency required by combining theorem.

Q15

Intermediate

3 marks - 2-var K-map

Draw/fill the 2-variable K-map for $F(A,B)=\sum m(0,2)$ and give the minimal SOP.

WORKED SOLUTION

	B=0	B=1
A=0	1	0
A=1	1	0

Group of 2 (vertical) → B'

FINAL ANSWER → $F = B'$

Q16

Intermediate

4 marks - 3-var K-map

Fill 3-var K-map for $F(A,B,C)=\Sigma m(0,2,3,5)$ and obtain minimal SOP.

WORKED SOLUTION

	00	01	11	10
0	1	0	1	1
1	0	1	0	0

Groups: $A'C'$ (m_0+m_2 wrap), $A'B$ (m_2+m_3), $AB'C$ (m_5)

FINAL ANSWER → $F = A'B + A'C' + AB'C$

Q17

Intermediate

3 marks - Legal Groups

Which groupings are legal on a 3-var K-map? (Select all)

A. Four corners B. 2×2 block C. Three adjacent 1s D. Two cells differing in two variables

WORKED SOLUTION

A legal (size 4 wrap). B legal. C illegal (size $\neq$ power of 2). D illegal (not adjacent).

FINAL ANSWER → A and B

Q18

Intermediate

3 marks - Variable Cancellation

On a 3-var K-map a group of four 1s covers cells m_0, m_2, m_4, m_6 . What is the product term?

WORKED SOLUTION

Cells: 000,010,100,110 → C is always 0; A and B change → term = C'

FINAL ANSWER → C'

Q19

Intermediate

3 marks - Group of 8

A 3-variable function has all eight minterms = 1. What is the minimal expression?

WORKED SOLUTION

The whole map is one group of 8 → $F = 1$.

FINAL ANSWER → $F = 1$

Q20

Intermediate

4 marks - Fill Empty Map

Given $F = A'C + AB$, place the 1s on a 3-var K-map and list the minterms.

WORKED SOLUTION

$A'C$ covers m1 (001) and m3 (011). AB covers m6 (110) and m7 (111). Minterms: 1,3,6,7

FINAL ANSWER → $\Sigma m(1,3,6,7)$

SECTION D

Advanced K-map Techniques

Intermediate → Advanced • Q21--Q26

ADVANCED K-MAP STRATEGY

- Wrap-around and four-corner groups are valid
- Don't-cares (X) treated as 1 only when they enlarge a group
- Prime implicant = largest group that cannot be grown further
- Essential PI covers at least one 1 not covered by any other PI
- Minimal cover = all EPIs + smallest set of remaining PIs

Q21

Advanced

5 marks - 4-var + Corners

$F(A,B,C,D) = \sum m(0,2,5,7,8,10,13,15)$. Obtain minimal SOP. Watch for corner grouping.

WORKED SOLUTION

AB\CD	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

Four corners → $B'D'$; quad m5,7,13,15 → BD

FINAL ANSWER → $F = B'D' + BD$

Trap: Missing the four-corner group forces a non-minimal expression.

Q22

Advanced

4 marks - Wrap-around

On a 4-var K-map the cells m0, m2, m8, m10 are 1. What single product term covers all four?

WORKED SOLUTION

These are the four corners. Variables B and D are 0 throughout → $B'D'$

FINAL ANSWER → $B'D'$

Q23

Advanced

6 marks - PI / EPI

For $F(A,B,C)=\Sigma m(0,1,2,5,6,7)$: list all PIs, identify EPIs, give one minimal cover.

WORKED SOLUTION

PIs: $A'B'$, $A'C'$, $B'C$, BC' , AC , AB

One minimal cover: $A'B' + BC' + AC$

FINAL ANSWER → $F = A'B' + BC' + AC$ **Q24**

Advanced

4 marks - Essential Count

A function has five prime implicants of which three are essential. What is the minimum number of product terms in any minimal SOP?

WORKED SOLUTION

All three EPIs must appear. If they already cover every 1, the minimum is exactly 3.

FINAL ANSWER → At least 3 (exactly 3 if EPIs cover all 1s)**Q25**

Advanced

4 marks - Overlapping Groups

True or False: Overlapping groups on a K-map are always illegal.

WORKED SOLUTION

False. Overlapping is legal and frequently necessary for largest groups and a minimal expression.

FINAL ANSWER → False**Q26**

Advanced

5 marks - Minimal Cover Selection

F has PIs: $P1=A'B$, $P2=BC'$, $P3=AC$, $P4=A'C'$. The 1s covered uniquely are: m_0 only by $P4$, m_7 only by $P3$. Select a minimal cover.

WORKED SOLUTION

EPIs are $P4$ and $P3$. Remaining 1s covered by choosing among $P1/P2$. One minimal cover: $A'C' + AC + BC'$ (or $+ A'B$).

FINAL ANSWER → Example: $A'C' + AC + BC'$

SECTION E

DeMorgan & Bubble Pushing

Intermediate • Q27--Q30

DeMORGAN / BUBBLE STRATEGY

- Outside-in: $(XY)' = X' + Y'$, $(X + Y)' = X'Y'$
- Never stop after one application if nested complements remain
- Bubble on output $\equiv$ bubbles on all inputs of the dual gate
- Two bubbles in series cancel

Q27

Intermediate

3 marks - Nested DeMorgan

Simplify $F = ((A+B)' \cdot C)' + (A \cdot B)'$. Show every step.

WORKED SOLUTION

$$(A+B)' = A'B', (AB)' = A' + B'$$

$$F = (A'B' \cdot C)' + (A' + B') = (A+B+C') + (A' + B') = 1$$

FINAL ANSWER $\rightarrow F = 1$

Trap: Stopping after first DeMorgan leaves nested complements.

Q28

Intermediate

3 marks - Bubble Push

Convert $(A \cdot B)' + C$ into simplest form with primary inputs.

WORKED SOLUTION

$$(AB)' = A' + B' \rightarrow F = A' + B' + C$$

FINAL ANSWER $\rightarrow F = A' + B' + C$

Q29

Intermediate

3 marks - Dual Form

Apply DeMorgan to convert $(A \cdot B \cdot C)'$ into an OR with inverted inputs.

WORKED SOLUTION

$$(ABC)' = A' + B' + C'$$

FINAL ANSWER $\rightarrow A' + B' + C'$

Q30

Intermediate

4 marks - Nested + Simplify

Simplify $F = ((A'+B)' + (A \cdot C'))'$.

WORKED SOLUTION

Inner: $(A'+B)' = AB'$, $(AC)' = A'+C'$

$F = (AB' + A' + C')' = (AB')' \cdot (A')' \cdot (C')' = (A'+B) \cdot A \cdot C = ABC$

FINAL ANSWER $\rightarrow F = ABC$

SECTION F

Universal Gates

*Advanced • Q31--Q33***UNIVERSAL GATE STRATEGY**

- Simplify the function first
- NOT = NAND/NOR with tied inputs
- AND = NAND + inverter (or DeMorgan form)
- OR = inverted inputs into NAND
- Count 2-input gates; multi-level may reduce total
- Verify by expanding back

Q31**Advanced**

3 marks - NAND Primitives

Show 2-input NAND realisations of NOT, AND and OR. State gate counts.

WORKED SOLUTION

NOT: A tied $\rightarrow$ 1 NAND

AND: NAND then NAND-inverter $\rightarrow$ 2 NANDs

OR: invert both inputs then NAND $\rightarrow$ 3 NANDs

FINAL ANSWER $\rightarrow$ NOT=1, AND=2, OR=3**Q32****Advanced**

3 marks - NOR Primitives

Show 2-input NOR realisations of NOT, OR and AND. State gate counts.

WORKED SOLUTION

NOT: A tied $\rightarrow$ 1 NOR

OR: NOR then NOR-inverter $\rightarrow$ 2 NORs

AND: invert inputs then NOR $\rightarrow$ 3 NORs

FINAL ANSWER $\rightarrow$ NOT=1, OR=2, AND=3

Q33**Advanced**

5 marks - NAND-only F

Implement $F = AB + C$ using only 2-input NAND gates. Minimise gate count.

WORKED SOLUTION

$F = ((AB)' \cdot C')' \rightarrow$ three NANDs: one for AB, one for C', one final.

FINAL ANSWER $\rightarrow$ 3 NAND gates

SECTION G

Exam / GATE Challenge

*Advanced • Q34--Q35***Q34**

GATE-Style

2 marks - MCQ

Minimum number of 2-input NAND gates to implement $F=(A+B)(C+D)$ is
A. 3 B. 4 C. 5 D. 6

WORKED SOLUTION

Classic multi-level NAND count for this function is 6.

FINAL ANSWER → D. 6**Q35**

GATE-Style / MSQ

3 marks - Concepts

Which statements are true?

- A. Number of PIs always equals number of EPIs
- B. Minimal SOP must contain every EPI
- C. Don't-care minterms may appear in a PI
- D. Two minimal covers can differ in number of product terms (classic definition)

WORKED SOLUTION

A False · B True · C True · D generally False under classic term-count minimality

FINAL ANSWER → B and C**Sources & Further Practice**

- GATE Overflow – Digital Logic / K-Maps archives
- ExamSIDE GATE ECE / CSE previous-year collections
- NPTEL Digital Circuits (IIT Madras, IIT Bombay)
- Morris Mano – Digital Design; Roth – Fundamentals of Logic Design

End of Lab 2

Fewer gates. Same truth table. Better design.

35 QUESTIONS • WORKED SOLUTIONS • PRINT READY